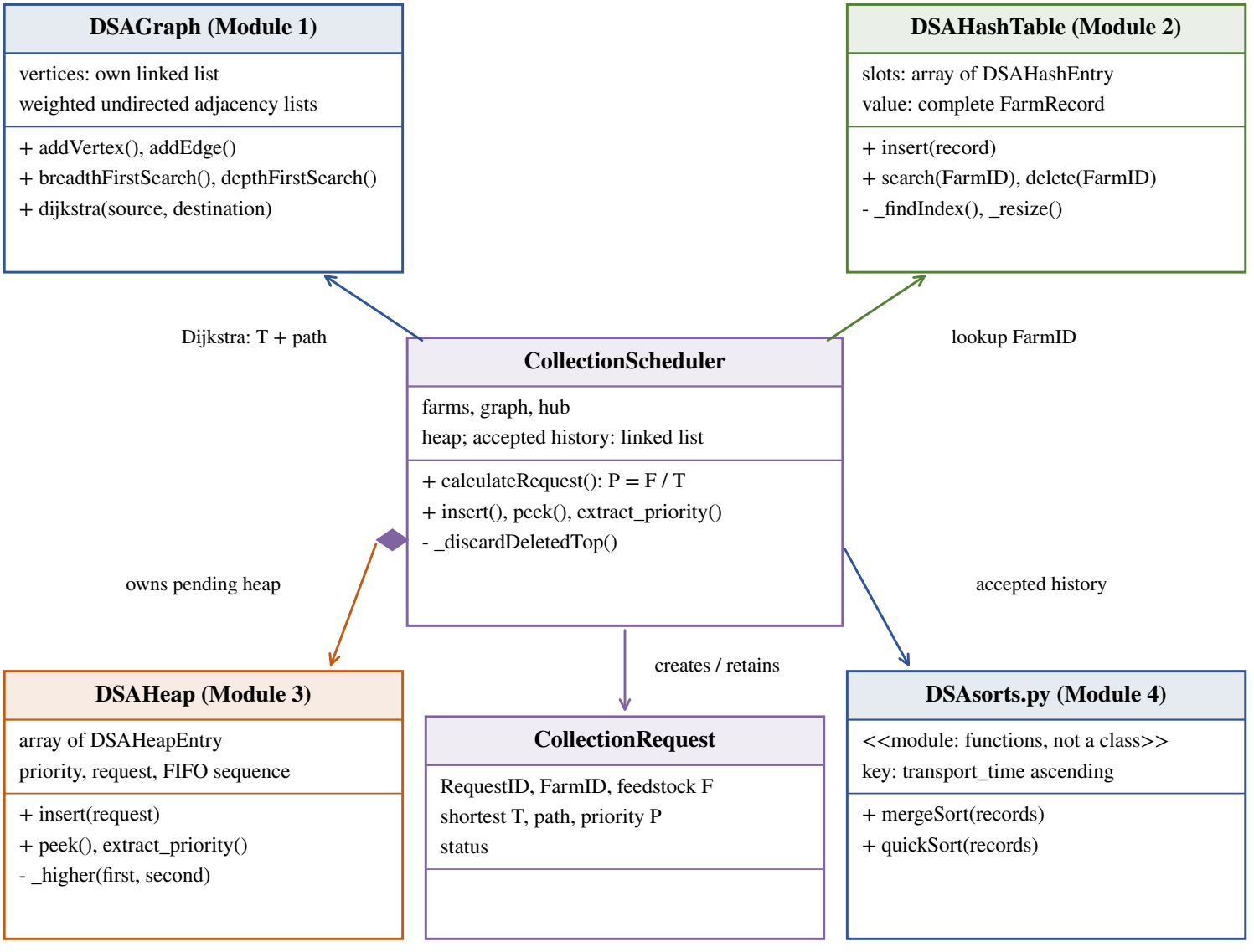
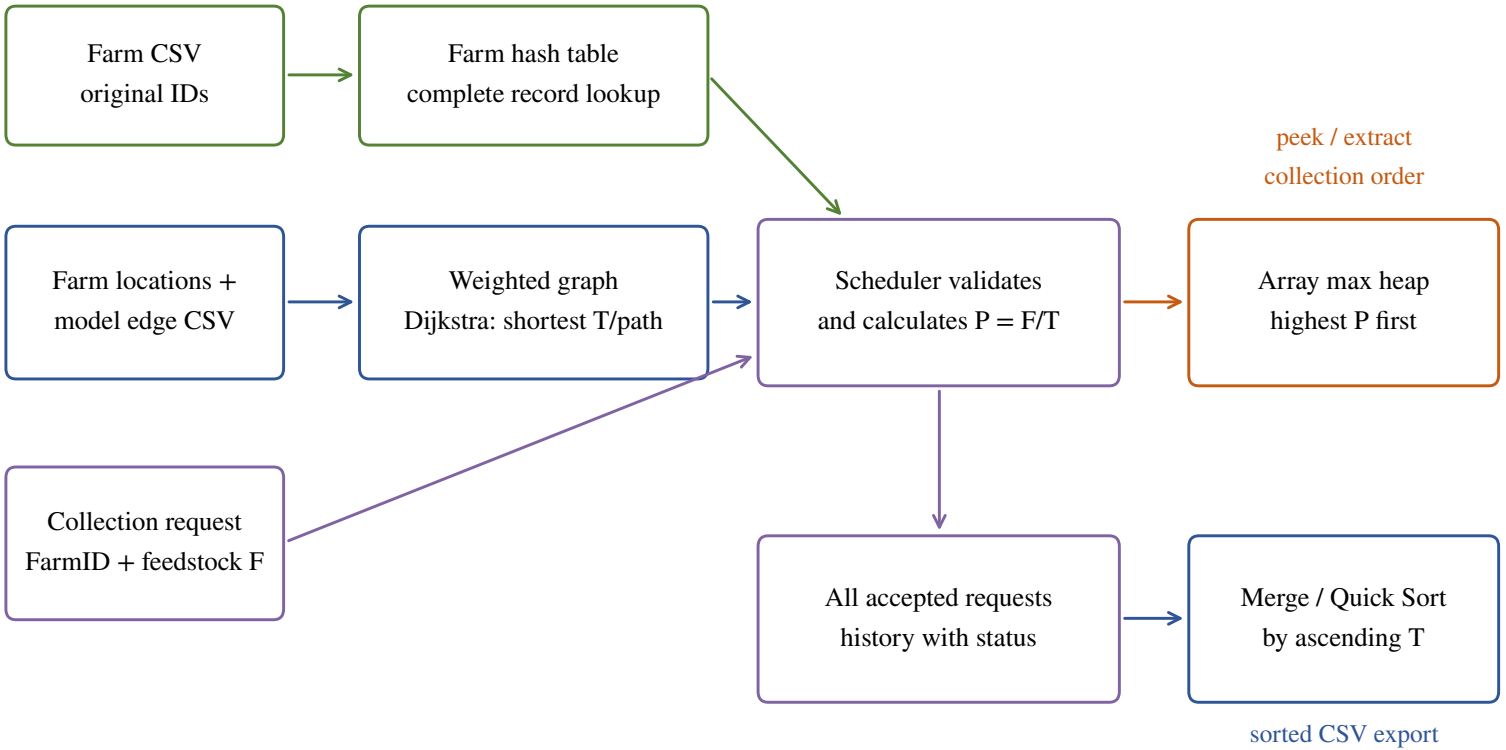
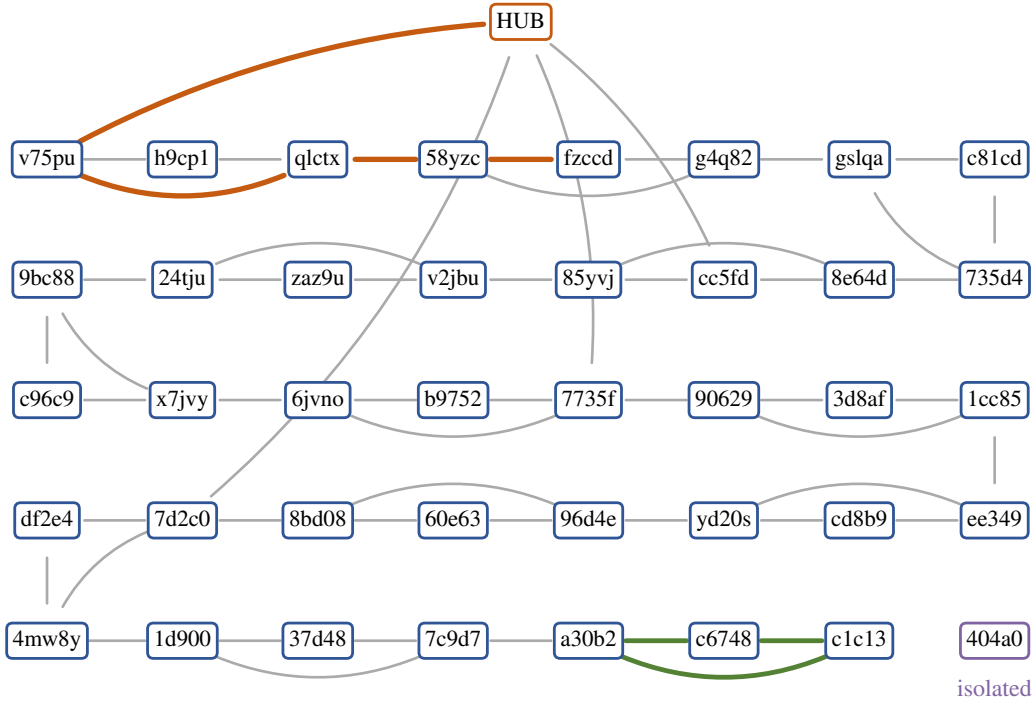
Implemented classes and four-module interaction



Class compartments show attributes and operations; arrows denote use/data dependencies.

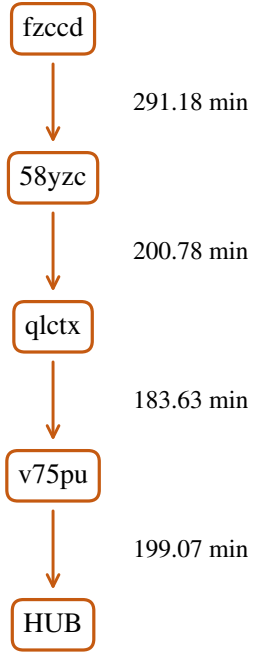

Data flow: lookup, routing, scheduling and reporting
Farm CSV
original IDs
Farm hash table
complete record lookup
peek / extract
collection order
Farm locations +
model edge CSV
Weighted graph
Dijkstra: shortest T/path
Scheduler validates
and calculates P = F/T
Array max heap
highest P first
Collection request
FarmID + feedstock F
All accepted requests
history with status
Merge / Quick Sort
by ascending T
sorted CSV export

(a) Full schematic: 41 vertices, 55 edges



Other connection R005 shortest route Detected cycle

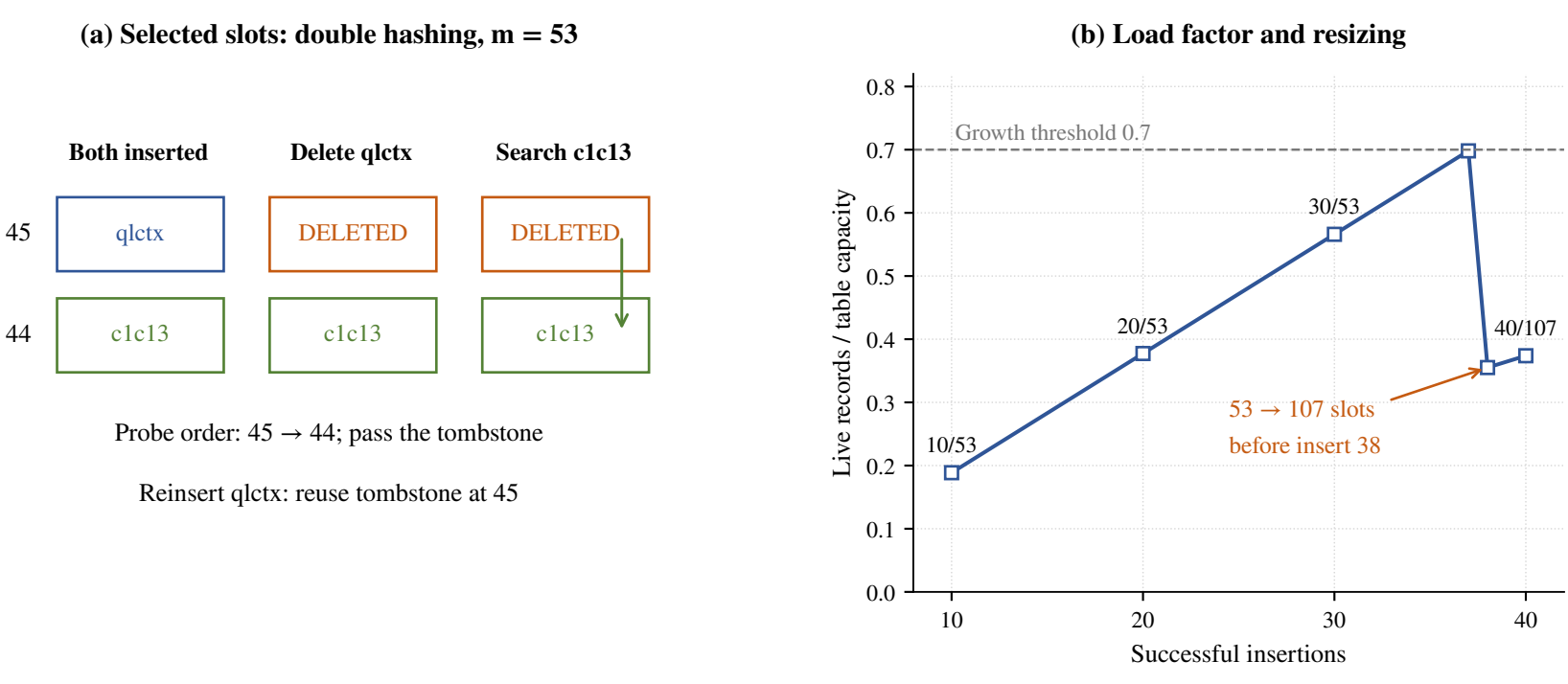
(b) R005 route to HUB



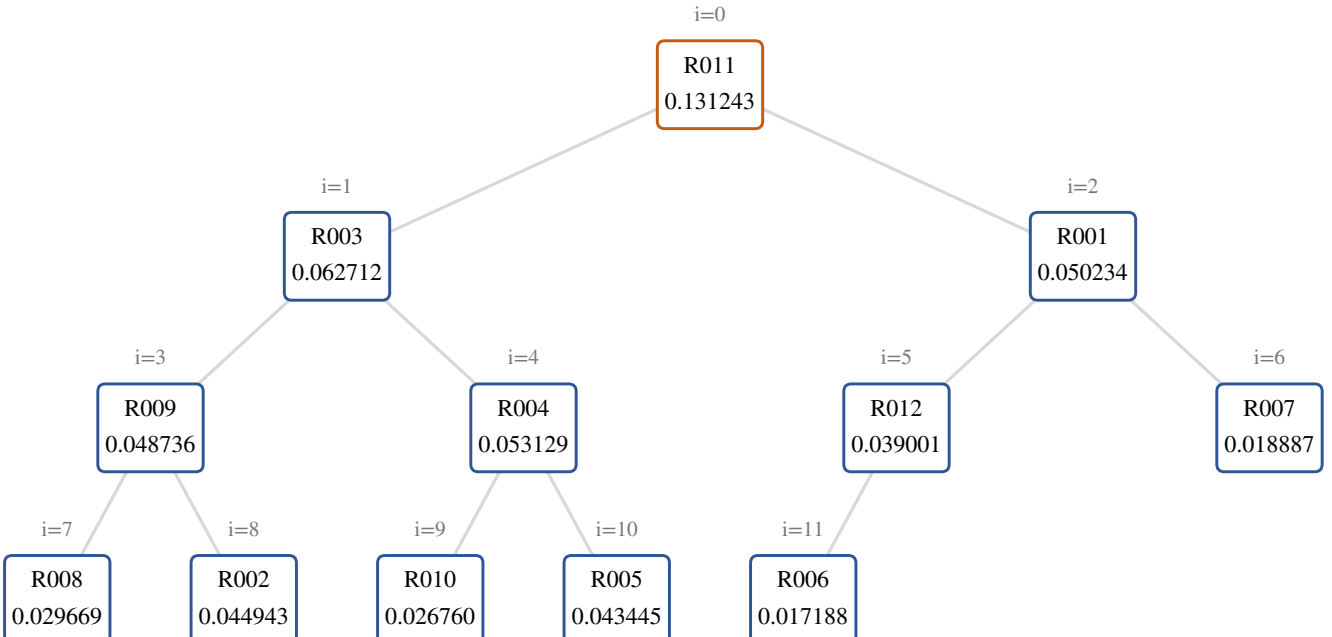
Shortest total: 874.66 min

BFS from HUB: minimum number of connections

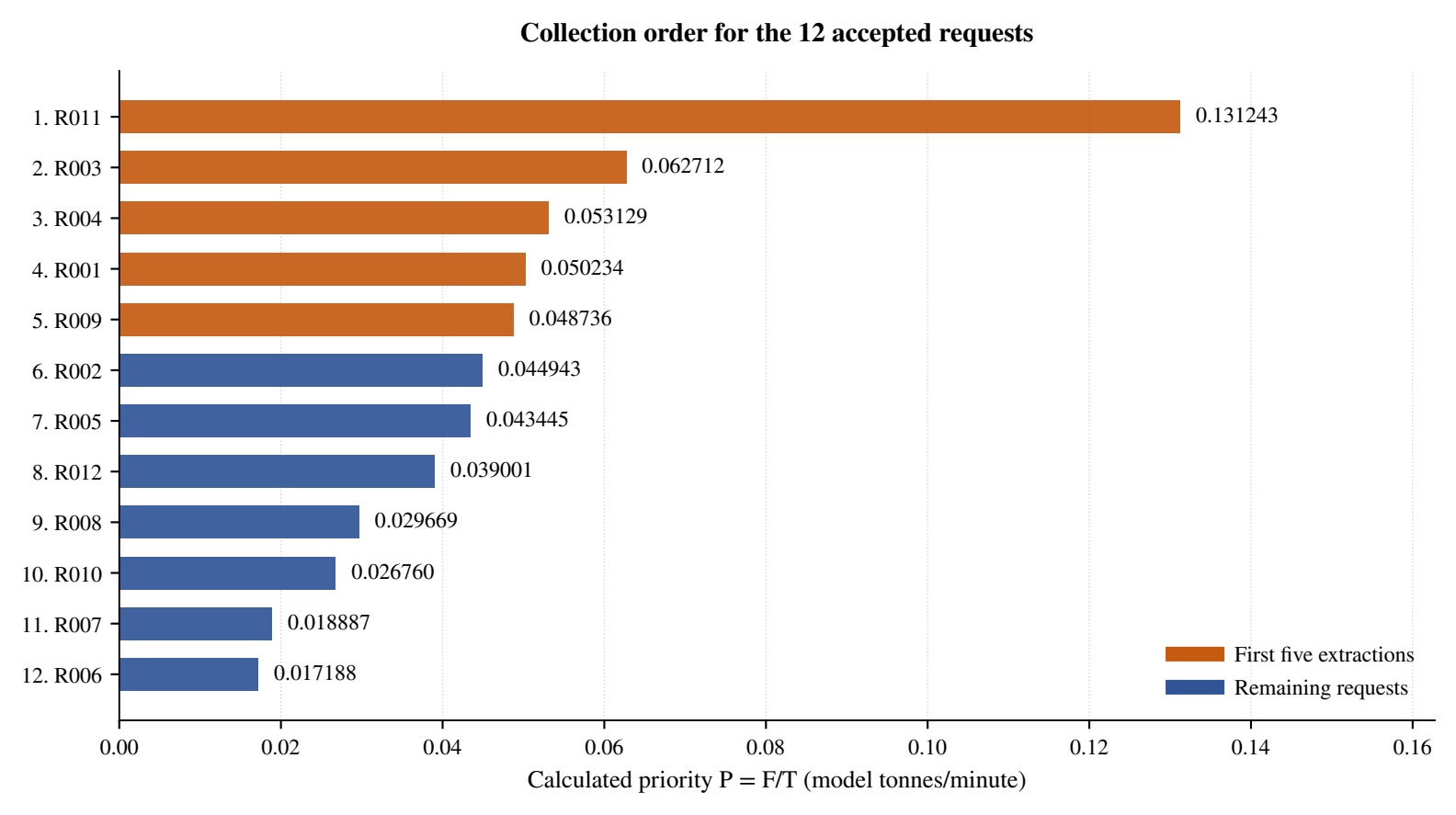
Level 0	HUB
Level 1	v75pu cc5fd 7735f 7d2c0
Level 2	h9cp1 qlctx 8e64d 85yvj b9752 90629 6jvno 8bd08 df2e4 4mw8y
Level 3	58yzc 735d4 v2jbu 3d8af 1cc85 x7jvy 60e63 96d4e 1d900
Level 4	fzccd g4q82 c81cd gslqa zaz9u 24tju ee349 c96c9 9bc88 yd20s 37d48 7c9d7
Level 5	cd8b9 a30b2
Level 6	c6748 c1c13
Unreachable	404a0 (isolated in the model)

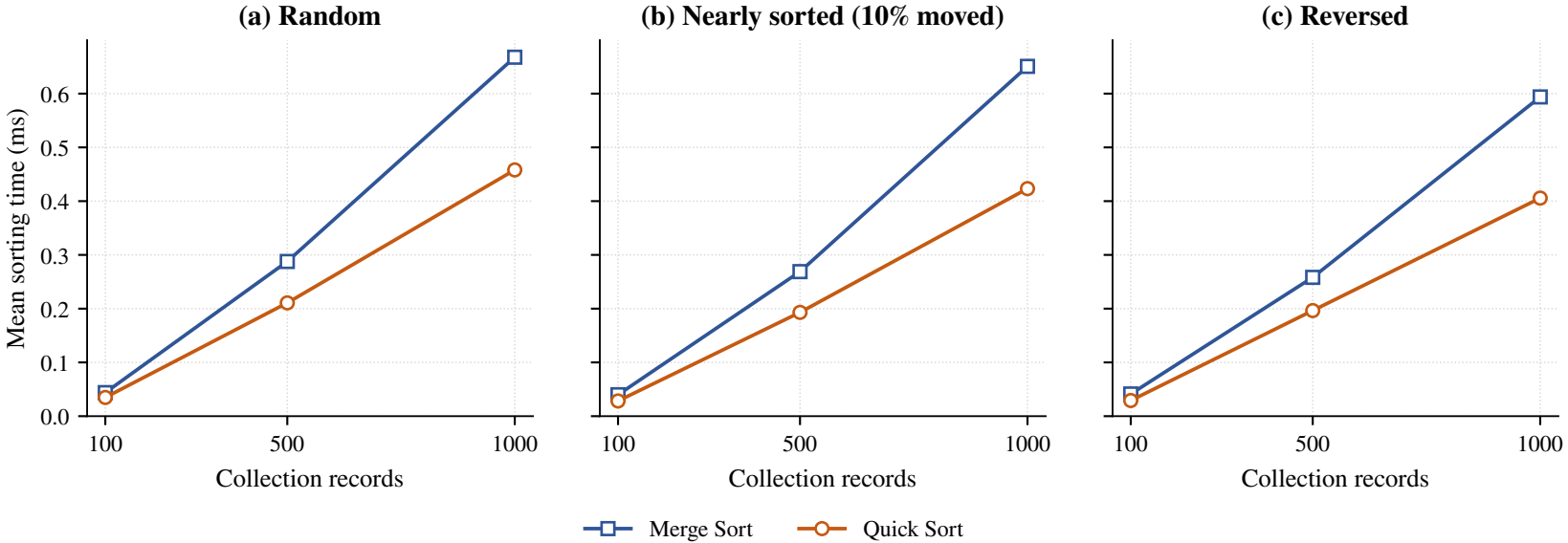

(a) Selected slots: double hashing, m = 53
Both inserted
Delete qlctx
Search c1c13
45
44
qlctx
c1c13
DELETED
DELETED
c1c13
c1c13
Probe order: 45 → 44; pass the tombstone
Reinsert qlctx: reuse tombstone at 45
(b) Load factor and resizing
Growth threshold 0.7
10/53
20/53
30/53
40/107
53 → 107 slots
before insert 38
Live records / table capacity
Successful insertions
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
10
20
30
40

Max heap after 12 accepted insertions



Labels: RequestID and P = F/T. Children of index i: 2i + 1 and 2i + 2.


Collection order for the 12 accepted requests
1. R011
2. R003
3. R004
4. R001
5. R009
6. R002
7. R005
8. R012
9. R008
10. R010
11. R007
12. R006
0.131243
0.062712
0.053129
0.050234
0.048736
0.044943
0.043445
0.039001
0.029669
0.026760
0.018887
0.017188
First five extractions
Remaining requests
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.16
Calculated priority P = F/T (model tonnes/minute)


(a) Random
(b) Nearly sorted (10% moved)
(c) Reversed
Mean sorting time (ms)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
100
500
1000
Collection records
Merge Sort
Quick Sort